

Algebra 1
Unit 2
Lesson 1 Homework

Name _____
Date _____
Period _____

1. The product of powers law of exponents states to keep the then _____ the powers.

☐ add
☐ subtract
☐ multiply

☐ base
☐ coefficient
☐ exponent

and

2. Jeffrey simplified the algebraic expression and made an error. Find and fix the mistake.

$$\begin{aligned} &(-2x^2y^3)^4(3.5x^0y) \\ &(-2x^8y^{12})(3.5y) \\ &(-2 \cdot 3.5)(x^8)(y^{12} \cdot y) \\ &-7.5x^8y^{13} \end{aligned}$$

3. Write a note to Jeffrey explaining the error he made in the previous question.

Determine an equivalent algebraic monomial expression for each expression using the laws of exponents. Show your work.

4. $(-4a^2b)(5ab^3)$

5. $(\frac{-1}{2}x^3y^0)(6x^2y^3)(-3y)$

$$6. \quad (5.6kj^8)(-3.25k^9)(\frac{1}{4}k^7j^{12})$$

$$7. \quad (-5mn^3p^5)^2$$

$$8. \quad (4.1c^{-4}d)(-2.2c^{10}d^0)(-c^{-2}d^3)$$

$$9. \quad (\frac{5}{8}r^{-2})^0(-7.6r^4s^5)^{-2}$$

$$10. \quad (-2y^3z)^5(6y^{10}z^8)^2$$

Algebra 1
Unit 2
Lesson 2 Homework

Name _____
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Period _____

1. The **quotient of powers law** of exponents states keep the then _____ the powers.

add
subtract
multiply

base
coefficient
exponent

2. The **power of a quotient law** of exponents states the exponential expression $(\frac{a}{b})^m$ is equal to

- ☐ a^{b-m} .
☐ $\frac{a^m}{b^m}$.
☐ $a^m - b^m$.

Find the quotient of each expression by using the laws of exponents to generate an equivalent expression.

3. $\frac{-4x^4y^2}{10xy^7}$

4. $\frac{3a^{10}b^6}{6b^2c}$

5. $(\frac{-2r^3s}{3t^4})^6$

6. $\frac{-k^7j^3}{7k^9j^3}$

$$7. \quad \frac{46w^3z^{10}}{33x^2y^{11}}$$

$$8. \quad \frac{-144c^9d^2e^0}{98c^4d^{11}e}$$

$$9. \quad \left(\frac{z^0}{-3x^3}\right)^4$$

$$10. \quad \frac{5s^{15}t^6}{-10s^{12}t}$$

1. Tamer was simplifying the expression and made an error. His steps are shown. Find and fix the mistake.

$$\left(\frac{-2xy \cdot 3x^2}{5x^4y}\right)^2$$

Step 1: $\left(\frac{-6x^3y}{5x^4y}\right)^2$

Step 2: $\frac{-6x^6y^2}{5x^8y^2}$

Step 3: $\frac{-6}{5x^2}$

2. Write a note to Tamer explaining his error.

For questions 3-5, find an equivalent expression using the laws of exponents.

3. $\frac{-6a^7b^9}{(-2ab^3)^5}$

4. $\frac{(9x^8y) \cdot (-2x^2y^5)}{3x^5y^{10}}$

5. $\left(\frac{-3mn^3}{4m^4 \cdot 2m^2n^5}\right)^3$

6. Prove $2^{3x+5} = 32(8)^x$ by using the laws of exponents.

7. Prove $27^{6a-1} = \frac{(3^{6a})^3}{9 \cdot 3}$ by using the laws of exponents.

8. Write an equivalent expression for $(-5.2)^{2x^3} \cdot (-5.2)^{x^2} \cdot (-5.2)^{-5}$.

9. Use the law of exponents to write an equivalent expression for 4^{3-2c} .

10. Use the law of exponents to write an equivalent expression for 3.5^{3x+7} .

Algebra 1

Unit 2

Lesson 4 Homework

Name _____

Date _____

Period _____

1. True or False? The polynomial $-7xy^4 + 4x^5y^2 - 8.6x^3y^3$ is written in standard form.

2. Write the polynomial $-7xy^4 + 4x^5y^2 - 8.6x^3y^3$ in standard form.

For questions 3-4, rewrite each polynomial in standard form, then identify the features of the polynomial.

3. $2.5y^3 + 7y - 8\frac{7}{8}y^5$

Standard Form _____

Number of Terms _____

Degree of the polynomial _____

Leading Term _____

Leading Coefficient _____

4. $-\frac{3}{4}a^4b^2 - 16ab^3 + ab^6$

Standard Form _____

Number of Terms _____

Degree of the polynomial _____

Leading Term _____

Leading Coefficient _____

5. Create a polynomial that has 4 terms, a degree of 6, a leading coefficient of -1 , and a constant term of -5.5 .
6. Samantha found the product of $-2g^3(\frac{1}{4}g^8h^3 + 5g^2h^6 - g^9)$ to be $-\frac{1}{2}g^{24}h^3 - 10g^6h^6 - 2g^{27}$. Do you agree or disagree with her? Explain.
7. Find the product of $-2g^3(\frac{1}{4}g^8h^3 + 5g^2h^6 - g^9)$ in standard form.

Find the product of each by distributing the monomial to the polynomial. Express the product in standard form.

8. $-5x^2(-3xy + 2x^7 - 7x^3)$

9. $(-\frac{5}{8}c^4d + 10c^5d^3 - cd^2)(\frac{3}{5}c^2)$

10. $-4.2a^3(2b^5 - 1.5a^2b - 9)$

Algebra 1

Unit 2

Lesson 5 Homework

Name_____

Date_____

Period_____

1. Are the terms $-4xy^2$ and $\frac{1}{2}x^2y$ like terms? Why or why not? Explain.

2. Find the sum of $(6 - t - t^2)$ and $(10t + t^2)$.

3. Write the sum from question 2 in standard form.

4. Find the difference of $(5.5n^2 + 8 - 3n)$ and $(-3n^2 - 5)$.

5. Write the difference from question 4 in standard form.

6. What polynomial must be added to $-5c^2 - 3 + 4c$ in order to get a sum of $-2c^2 - c + 1$?

For questions 7-10, find each sum or difference. Write the result in standard form.

7. $(-3.4g^2h - 5gh^2 + 7) + (-1.75gh^2 - 12)$

8. $2\frac{1}{5}r^5t^2 - (3r^4t^3 + r^5t^2 - 6rt^4)$

9. $(8a^2 - 3a + 11b^2) + (2a - 5b^2 - 2a^2)$

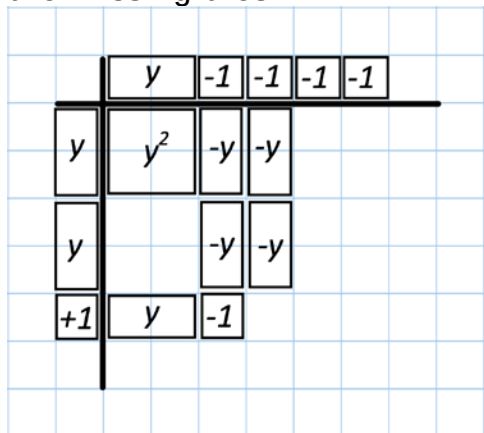
10. $(9.3m^2 - 5.1n) - (2n^2 + 6.75m)$

1. Model the following expression with algebra tiles.

$$5x - 4$$

Use the expression $(2y + 1)(y - 4)$ for questions 2 – 4.

2. Complete the model to find the product of $(2y + 1)(y - 4)$ by drawing and labeling the missing tiles.



3. Determine how many of each tile are represented in the product and write the numbers in the table.

y^2 tiles	$+y$ tiles	$-y$ tiles	$+1$ tiles	-1 tiles

4. Complete the statement.

The product of $(2y + 1)(y - 4)$ is _____.

For questions 5-9, use algebra tiles to find each product.

5. $(4 - 3x)(x - 1) =$ _____



6. $(2x - 1)^2 =$ _____



7. $(x - 3)(x + 3) =$ _____



8. $(x - 5)(x + 1) =$ _____



9. $(1 - x)(2x + 2) =$ _____



10. One of your classmates was absent when you learned how to multiply polynomials using algebra tiles. Write them a note explaining the process.

Algebra 1

Unit 2

Lesson 7 Homework

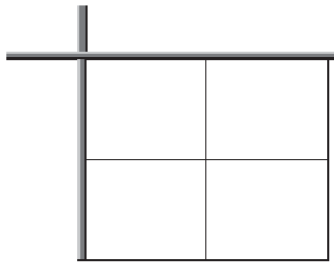
Name _____

Date _____

Period _____

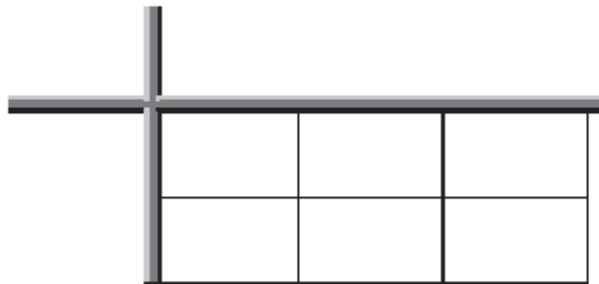
1. Find the product of $(x^2 - 3)(4x + 2)$ using the distributive property.

2. Find the product of $(x^2 - 3)(4x + 2)$ using an area model.



3. Find the product of $(-\frac{1}{4}y + 3y^2 - 7)(2y - 5)$ using the distributive property.

4. Find the product of $(-\frac{1}{4}y + 3y^2 - 7)(2y - 5)$ using an area model.



Find each product.

5. $(-6a + \frac{2}{5}b)^2$

6. $(2.4 - 3c)(-0.3 + 5c^3 - c^2)$

7. $(\frac{3}{4}x - 7)(\frac{-2}{5} + 8x)$

8. $(-2\frac{1}{2} + 5d^2)(6d - 6.1c - 3d^2)$

9. $(12 + 6m)(-8n - \frac{18}{5})$

10. $(4r^2 - \frac{7}{4} + 19r)(-3r + 2r^2 + 15)$

Algebra 1
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Lesson 8 Homework

Name _____
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1. Rewrite the expression in each step using the description: $-2(x - 2) + 3x$

	Distribute the -2 to $(x - 2)$.
	Combine like terms.

2. Rewrite the expression in each step using the description: $5x + 6(2x + 7)$

	Distribute the 6 to $(2x + 7)$.
	Combine like terms.

Use the order of operations to simplify each polynomial.

3. $4(x - 3) + 6(x - 2)$

4. $\frac{2}{3}(x - 3) + 6x$

5. $2(x^2 - 3x) + 4(x + 5)$

6. $4x^2 - 4(x - 8)$

7. $(x + 2)(x - 3) - 6(x^2 + x)$

8. $-\frac{2}{5}(5x - 15) + 5x$

9. $x^2 + (x - 2)(x - 2)$

10. $4x(x + 1) + 2(x + 5)$

1. Use the area model to find $(-4.5x^4y^3 + 2.4x^3y^2 - 3.6x^2y) \div (3xy)$ where $x \neq 0$ and $y \neq 0$.

Find each quotient.

2. $(\frac{3}{4}x^3 + \frac{1}{2}x^2) \div (-\frac{1}{2}x^2)$ where $x \neq 0$.

3. $(4a^3 + 8a^2 - 20a) \div (4a)$ where $a \neq 0$

4. $(-10k^6r^4 + 6k^2r^4 - 42k^3r^3) \div (2k^2r)$ where $k \neq 0$ and $r \neq 0$

5. $(-17w^6 + 13w^3 - 18w^2) \div (2w)$ where $w \neq 0$

6. $\frac{12x^4-30x^3+6x^2-18}{6x}$ where $x \neq 0$

7. $\frac{14p^4+42p^3+21p^2-7p}{7p}$ where $p \neq 0$

8. $\frac{2t^4-2t^3+12t^2-7t}{2t}$ where $t \neq 0$

9. $\frac{2.8x^3+4.8+3.2x^2-0.8}{0.4x}$ where $x \neq 0$

10. $\frac{15y^3+45y^2+35y^2}{-5y^2}$ where $y \neq 0$